YUEYUE SHEN（沈越月）

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I'm a PhD student in astrophysics working on young star clusters. My research combines Gaia astrometry and spectroscopic surveys to investigate their stellar mass functions, chemical properties, and dynamical structure, as well as multiple-cluster systems. I am interested in understanding how stars and star clusters form and evolve in different Galactic environments.

EDUCATION

- **National Astronomical Observatories of China (NAOC), CAS** — Beijing, China
  Ph.D Student, in Astrophysics — *Sep 2022 - Present*
  - ***Supervisor:*** *Prof. Chao Liu*
- **Astrophysics and Space Science Observatory of Bologna (OAS), INAF** — Bologna, Italy
  Visiting Ph.D Student (CSC Scholarship) — *Sep 2025 - Sep 2026*
  - ***Supervisors:*** *Dr. Angela Bragaglia, Dr. Emanuele Dalessandro*
- **Zhejiang University of Technology** — Hangzhou, China
  Bachelor of Engineering, in Computer Science — *Sep 2018 - Jun 2022*
  - ***Supervisor:*** *Dr. Libo Weng, Prof. A-Li Luo (NAOC)*

RESEARCH EXPERIENCE

- **National Astronomical Observatories of China (NAOC), CAS** — Beijing, China
  Ph.D Research — *Sep 2022 - Present*
  - **The Stellar Initial Mass Function and Early Evolution of Young Stellar Systems**
    Supervisor: *Prof. Chao Liu*
    - Developed an IMF-aware N-body forward-modelling framework for 35 nearby young open clusters, demonstrating how the assumed IMF affects cluster evolution and magnitude-dependent membership completeness and contamination *(Shen et al., under revision at MNRAS)*.
    - Developed a Bayesian framework to infer present-day mass functions, ages, extinction and unresolved binary properties from Gaia colour–magnitude diagrams, accounting for membership selection effects and parameter degeneracies *(Shen et al., in prep.)*. - STARCAT
- **Astrophysics and Space Science Observatory of Bologna (OAS), INAF** — Bologna, Italy
  Visiting Ph.D Student Research — *Sep 2025 - Sep 2026*
  - **Probing Young Galactic Star Clusters: Metallicities and Kinematics with LAMOST and Gaia**
    Supervisor: *Dr. Angela Bragaglia*
    Combined LAMOST DR13 and Gaia DR3 to derive metallicities for 909 young clusters and radial velocities for 1,090 clusters, measure the Galactic radial metallicity gradient and identify 55 candidate cluster–cloud associations *(Shen et al., submitted to A&A)*.
  - **LISCA Systems in the Milky Way: Tracing the Hierarchical Assembly of Star Clusters**
    Supervisor: *Dr. Emanuele Dalessandro*
    Identified 356 candidate multi-cluster systems comprising 1,118 clusters in Gaia DR3. Age and richness trends are consistent with a hierarchical cluster assembly picture in which initially rich systems progressively lose substructure through dynamical evolution *(Shen et al., in prep.)*.
- **National Astronomical Observatories of China (NAOC), CAS** — Beijing, China
  Previous Research — *Mar 2022 - Sep 2024*
  - **Distance and Stellar Parameter of solar-like stars from LAMOST**
    Supervisor: *Prof. A-Li Luo*
    Derive spectroscopic distances and stellar parameters for half a million stars from LAMOST DR9 spectra using neural networks *(Shen & Luo, 2024)*. - SolarDis

Publications

4. **Probing Young Galactic Star Clusters: Metallicities and Kinematics with LAMOST and Gaia**
   Yue-Yue Shen *et al., submitted to A&A*
3. **IMF-aware N-body forward modelling of membership selection functions for young open clusters**
   Yue-Yue Shen *et al., under revision at MNRAS*
2. **Distance and stellar parameter estimations of solar-like stars from the LAMOST spectroscopic survey**
   Yue-Yue Shen *and A-Li Luo, A&A, 691, A218 (2024)*
1. **Constraints on the evolution of the Triton atmosphere from occultations: 1989–2022**
   Bruno Sicardy et al. (incl. ***Y.-Y. Shen****), A&A, 682, L24 (2024)*

Additional Experience

Teaching Assistant

- **Astrostatistics and Numerical Analysis** *Sep 2024 - Dec 2024*
  Graduate course at University of Chinese Academy of Sciences, Prof. Chao Liu

Observing Experience

- **2.16m telescope at Xinglong Observatory, Chengde, China** *6 nights, 2023 - 2024*
  Spectroscopic observations
- **0.7m telescope at Yanqi Lake Observatory, Beijing, China** *6 Oct 2022*
  Occultation light curve, contributed to *Bruno Sicardy et al. 2024*

Skills

- **Programming and Computation:** Python, C/C++, Java, SQL, HTML/CSS, LaTeX, Shell, Git
- **Statistics and Machine Learning:** Bayesian inference, Markov chain Monte Carlo (MCMC), Regression, Clustering, Classification, Neural Networks, Deep Learning

Awards and Honors

- **Excellent Student Prize** of University of Chinese Academy of Sciences *2024*
- **Academic Scholarship** of Zhejiang University of Technology *2021*

Conferences, Workshops and Seminars

- **High-resolution spectroscopy of open clusters, SPA-OC2 Workshop** Rome, Italy
  Contributed talk, "Investigating the Stellar Mass Function and Metallicity in Young Open Clusters" 10 –12 Dec 2025
- **Astrophysics Talk at Seminars of OAS, INAF** Bologna, Italy
  Seminar, recorded - ▶ *YouTube video* *11 Nov 2025*
- **Chinese Space Station Telescope(CSST) 2024 Annual Science Conference** HangZhou, China
  Participant *13 –18 May 2024*
- **Chinese Astronomical Society 2023 Annual Meeting** Weihai, China
  Participant *30 Aug –2 Sep 2023*